

Label-efficient Land Cover Classification at High Spatial Resolution using Multi-stage Pre-training

Dakota Hester, Vitor Souza Martins, Lucas Borges Ferreira

Mississippi State University
Department of Agricultural and Biological Engineering

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Land Cover

- **Land cover:** biophysical material covering the Earth's surface (e.g., water, vegetation, soil, structures, etc.) [1] **“What is there?”**

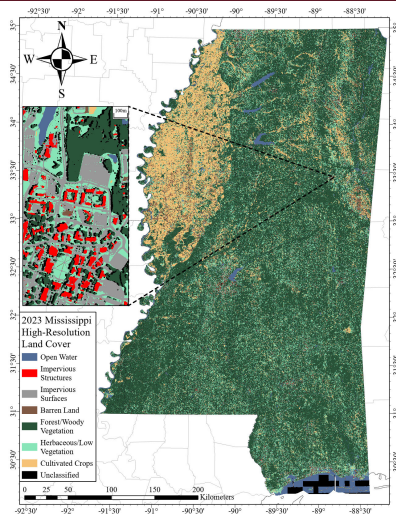


Figure 1: Land cover map of Mississippi.

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Land Cover

- **Land cover:** biophysical material covering the Earth's surface (e.g., water, vegetation, soil, structures, etc.) [1] **“What is there?”**
- Land cover data is used in a wide range of disciplines, including agriculture, forestry, urban planning, and environmental science [2–4]

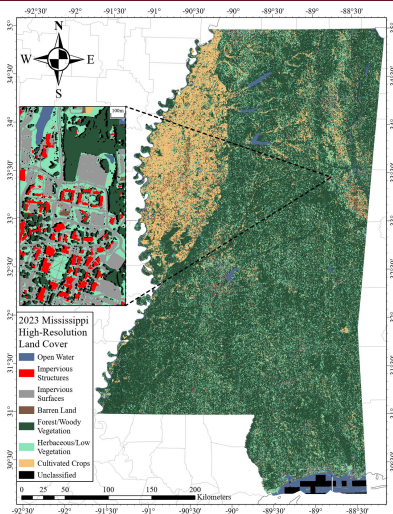


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- Prior to the advent of GIS, LULC data was human-interpreted from aerial photography [5, 6].

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Classifying LULC

- Prior to the advent of GIS, LULC data was human-interpreted from aerial photography [5, 6].
- With advent of satellite remote sensing in 1980s, researchers and practitioners began using automated algorithms to classify LULC [7].

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Classifying LULC

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- With advent of satellite remote sensing in 1980s, researchers and practitioners began using automated algorithms to classify LULC [7].
- Most classical approaches to LULC classification paradigms were pixel-wise statistical models that used coarse resolution multispectral data [8, 9].

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- Traditional land cover classifiers usually operate by transforming a feature vector into output class probabilities for a given pixel.

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- Common data sources include:

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Pixel-wise Classification

- Traditional land cover classifiers usually operate by transforming a feature vector into output class probabilities for a given pixel.
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- Traditional land cover classifiers usually operate by transforming a feature vector into output class probabilities for a given pixel.
- Common data sources include:
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- Common data sources include:
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 - Ancillary information (i.e., topography, soil, etc) [4, 10]

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Pixel-wise Classification

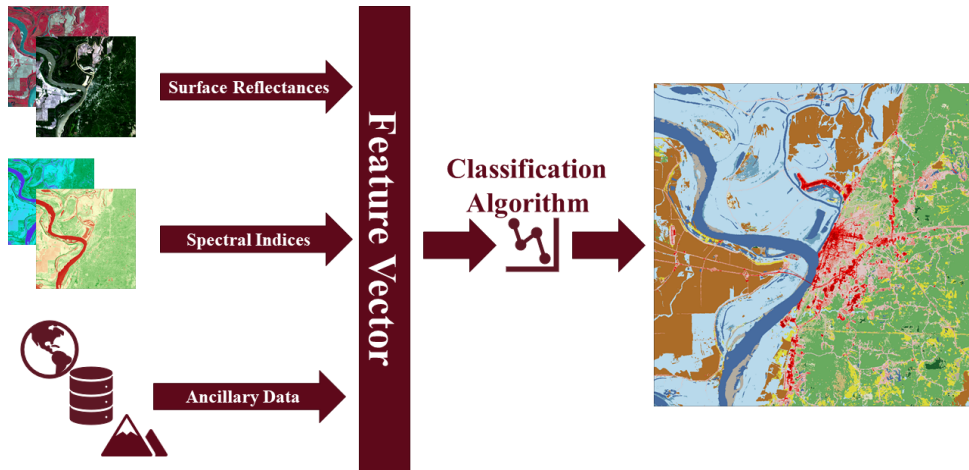


Figure 2: Pixel-wise classification workflow.

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Land Cover at High Spatial Resolution

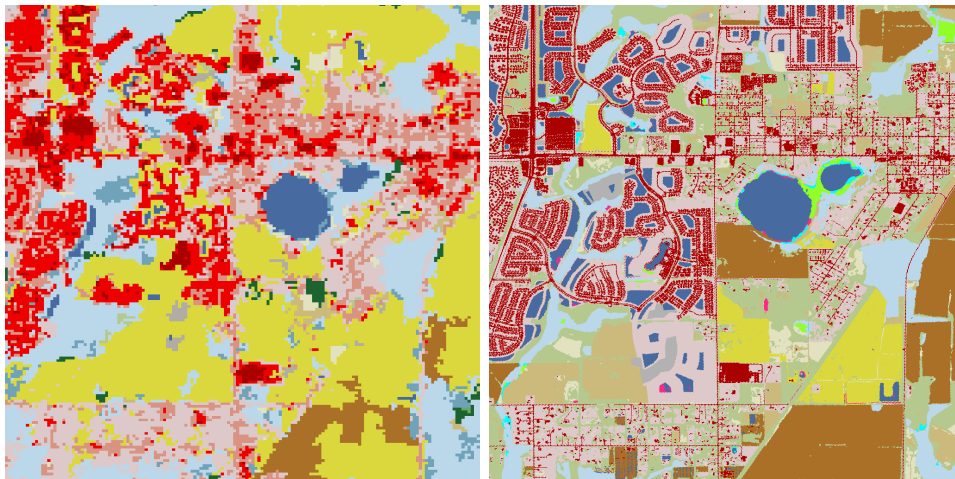


Figure 3: Comparison of land cover maps at 30 m (left) and 1 m (right) spatial resolution.

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Decreased Class Discriminability

Classifying pixels gets **more difficult** as spatial resolution **increases** [11, 12].

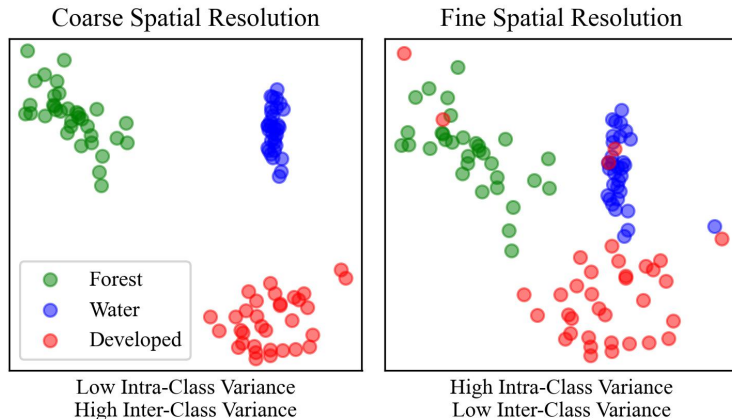


Figure 4: As spatial resolution increases, inter-class variance decreases [12].

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images/noisy_landcover.png

Object-Based Image Analysis (OBIA)

- One approach to classifying high spatial resolution data is to use **segmentation** to group neighboring pixels that share similar spectral characteristics into **objects**.

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- Instead of classifying individual pixels, the classifier operates on the objects.

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Object-Based Image Analysis (OBIA)

- One approach to classifying high spatial resolution data is to use **segmentation** to group neighboring pixels that share similar spectral characteristics into **objects**.
- Instead of classifying individual pixels, the classifier operates on the objects.
- Object-based image analysis (OBIA) addresses local variance concerns and produces cleaner, more accurate classification maps [13, 14].

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Object-Based Image Analysis (OBIA)

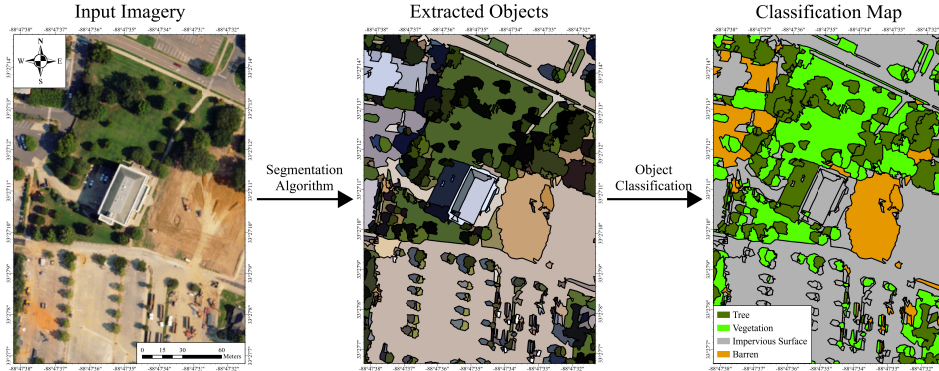


Figure 6: OBIA workflow.

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- OBIA is more complex, and requires more human supervision than pixel-wise classification methods [15].

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- OBIA is more complex, and requires more human supervision that pixel-wise classification methods [15].
- Ideally, we would like to have a generalizable method that can produce accurate classification maps without the need for extensive human intervention.

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“Why do we need deep models?”

- Deep learning models have several key advantages over traditional classifiers:
 - ① **Feature extraction** — deep models can learn complex hierarchies of **spatial** features that are difficult to engineer by hand [16].

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 - ① **Feature extraction** — deep models can learn complex hierarchies of **spatial** features that are difficult to engineer by hand [16].
 - ② **Flexibility** — the nature of deep learning architectures allows them to be applied to a wide range of tasks across different modalities, and can be modified to best suit a given task [17].

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 - ① **Feature extraction** — deep models can learn complex hierarchies of **spatial** features that are difficult to engineer by hand [16].
 - ② **Flexibility** — the nature of deep learning architectures allows them to be applied to a wide range of tasks across different modalities, and can be modified to best suit a given task [17].
 - ③ **End-to-end learning** — unlike traditional methods, deep models can be trained in an end-to-end fashion, where little to no pre-processing or post-processing is required [15, 18–20].

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- Convolutional filters emphasize local spatial patterns images.

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Spatial Feature Extraction

- Convolutional filters emphasize local spatial patterns images.
- A small filter moves across input imagery, performing a dot product with the input regions it covers at each location.

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- Convolutional filters emphasize local spatial patterns images.
- A small filter moves across input imagery, performing a dot product with the input regions it covers at each location.
- The result is a **feature map** that highlights the presence of certain spatial patterns at different locations in the input image.

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- Convolutional filters emphasize local spatial patterns images.
- A small filter moves across input imagery, performing a dot product with the input regions it covers at each location.
- The result is a **feature map** that highlights the presence of certain spatial patterns at different locations in the input image.
- The behavior of the filter is determined by its size, stride, and its parameters.

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Convolutional Filter

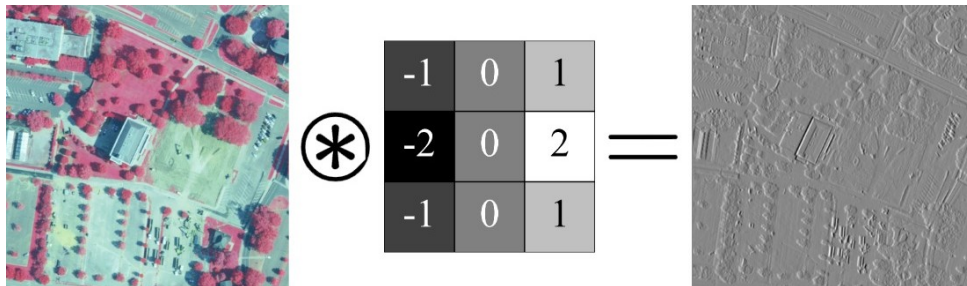


Figure 7: A convolutional operation on an aerial image. This convolutional filter highlights vertical lines.

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Convolutional Neural Networks (CNNs)

- Multiple convolutional filters can be applied to an input to extract feature maps that highlight different spatial patterns.

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Convolutional Neural Networks (CNNs)

- Multiple convolutional filters can be applied to an input to extract feature maps that highlight different spatial patterns.
- These feature maps can then be passed to more convolutional filters to extract more complex patterns.

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Convolutional Neural Networks (CNNs)

- Multiple convolutional filters can be applied to an input to extract feature maps that highlight different spatial patterns.
- These feature maps can then be passed to more convolutional filters to extract more complex patterns.
- The parameters of the convolutional filters are learned through an iterative process, where the gradient of a loss function that measures the difference between the predicted output and the true output with respect to the parameters is used to update the parameters [21, 22].

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Convolutional Neural Networks (CNNs)

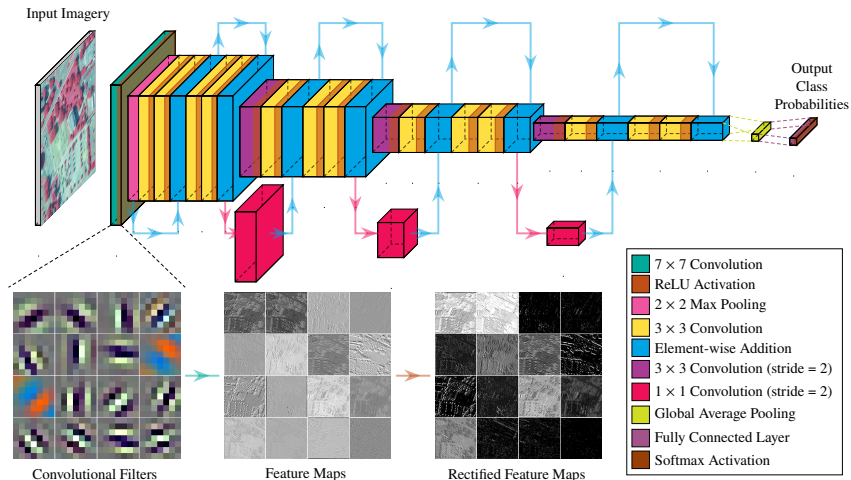


Figure 8: A ResNet-18 CNN architecture [23]. A sampling of learned convolutional filters from the first layer is shown.

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- A naive approach to land cover classification would be to assign a single label to each tile that is passed through a CNN, where this label represents the dominant land cover class in the tile [24–26].

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- A naive approach to land cover classification would be to assign a single label to each tile that is passed through a CNN, where this label represents the dominant land cover class in the tile [24–26].
- However, feature maps learned by a CNN are inherently **spatially structured**, and these feature maps can be leveraged to produce dense, pixel-wise classification maps through a process called **semantic segmentation**.

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- In an encoder-decoder architecture, an **encoder** CNN extracts features from the input such that a **decoder** network can upsample the features to the original input resolution [27].

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- The decoder network learns to generate a pixel-wise classification map from the features extracted by the encoder.

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- In an encoder-decoder architecture, an **encoder** CNN extracts features from the input such that a **decoder** network can upsample the features to the original input resolution [27].
- The decoder network learns to generate a pixel-wise classification map from the features extracted by the encoder.
- The most popular encoder-decoder network architecture is U-Net[28].

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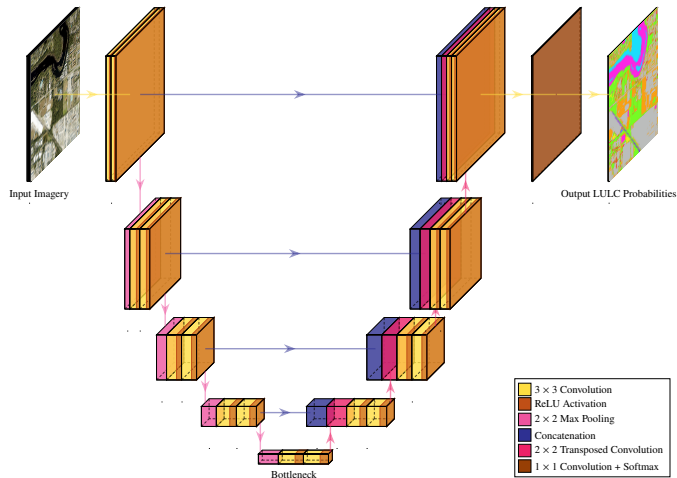


Figure 9: U-Net architecture [28].

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- While deep learning networks are powerful, their primary drawback is the amount of data required to train them.

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- While deep learning networks are powerful, their primary drawback is the amount of data required to train them.
- For example, ResNet-50, a popular CNN architecture, has over 25,000,000 parameters that need to be learned [23].

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Data Requirements

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- For example, ResNet-50, a popular CNN architecture, has over 25,000,000 parameters that need to be learned [23].
- For land cover, this is especially problematic, as high resolution data is complex and difficult to annotate at scale [29].

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- For land cover, this is especially problematic, as high resolution data is complex and difficult to annotate at scale [29].
- One approach to mitigate this issue is to use **transfer learning**, where a network can be **pre-trained** on a related task with a large amount of data, then **fine-tuned** on the target task with a smaller amount of data [30].

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Transfer Learning

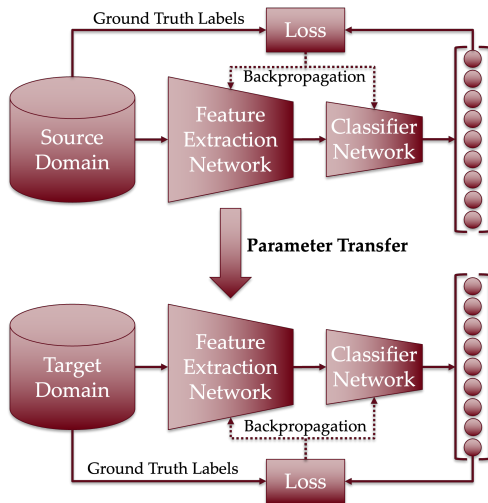


Figure 10: Transfer learning paradigm.

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- While labeled data is optimal for pre-training, some pre-training tasks can be performed in a **self-supervised** manner, where the network learns to predict some property of the input data without requiring labels [31].

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Denoising Autoencoder

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- Denoising autoencoders are a type of CNN that are trained to remove noise from an input image [32].

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- Denoising autoencoders are a type of CNN that are trained to remove noise from an input image [32].
- In effect, noise is added to an input image, then train the network to predict the original image from the noisy image.
- The network needs to learn **features** that are **robust** to noise while being meaningful for reconstruction [33].

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- Our overarching goal is to develop a 1m resolution land cover map that covers the entire state of Mississippi with a high degree of accuracy.

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- We construct a large U-Net model that undergoes various pre-training regimens, then fine-tune the model on only **119** tiles of fully annotated data from the state of Mississippi.

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- Data annotation is expensive and time-consuming, so the main focus of our work is to devise a creative pre-training strategy that combines self-supervised pre-training with a supervised fine-tuning stage to reduce the amount of labeled data required for training a deep model.
- We construct a large U-Net model that undergoes various pre-training regimens, then fine-tune the model on only **119** tiles of fully annotated data from the state of Mississippi.
- Transfer learning is not an uncommon approach, but self-supervised pre-training is not common in the remote sensing community — especially in the context of land cover classification. As such, we are most interested in the performance of our self-supervised model.

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- Our U-Net architecture is based on the original U-Net architecture proposed by Ronneberger, et al. [28].

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- Our U-Net architecture is based on the original U-Net architecture proposed by Ronneberger, et al. [28].
- In place of the original encoder network, we use a ResNet-152 architecture [23].

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- Our U-Net architecture is based on the original U-Net architecture proposed by Ronneberger, et al. [28].
- In place of the original encoder network, we use a ResNet-152 architecture [23].
- Overall, our complete U-Net architecture has over 100,000,000 parameters, making it a powerful model, but also difficult to train — especially in the absence of large amounts of data.

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- The ResNet-152 model has already been pre-trained on ImageNet, a large corpus of “natural” images.

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- The ResNet-152 model has already been pre-trained on ImageNet, a large corpus of “natural” images.
- We append a lightweight decoder network to the ResNet-152 encoder to form a denoising autoencoder. Training is performed using unannotated data over the state of Mississippi.

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- We then place the encoder in a U-Net architecture and train the network on a moderate amount of labeled data from another region where high-resolution data is available.
- Finally, the U-Net is fine-tuned on our limited amount of labeled data from Mississippi.

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USDA National Agricultural Imagery Program

- USDA NAIP imagery covers CONUS with 3-year revisit cycle at 0.6 m spatial resolution.

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 - Color infrared composites (Red, Green, and NIR bands) resampled to 1m spatial resolution
 - Imagery is clipped into tiles of 256x256 pixels

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 - 20% of tiles are placed in the pre-training set, while 5% in pre-training validation set (377,913 and 94,480 tiles, respectively)

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 - Imagery is clipped into tiles of 256x256 pixels
 - 20% of tiles are placed in the pre-training set, while 5% in pre-training validation set (377,913 and 94,480 tiles, respectively)
 - For fine-tuning, 250 tiles total were selected to be labeled (119 in training set, 61 in validation set, and 70 in test set)

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Chesapeake Bay Land Cover Data

- In order to enhance the model's ability to fit the data, we introduce a second pre-training stage using fully annotated data from the Chesapeake Bay Land Cover dataset.

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- 2018 NAIP tiles and corresponding land cover data extracted from Fairfax County, VA.

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- Why not directly apply these tiles to the fine-tuning stage?

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 - Generalization of the model to Mississippi is not guaranteed with data from Chesapeake Bay Watershed.
- 10,220 tiles used for training and 5,110 for validation.

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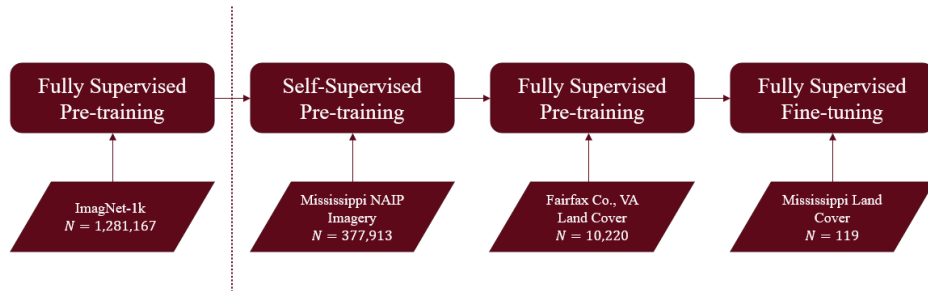


Figure 11: Model training flowchart.

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Pre-training Strategies

Pre-training with a denoising autoencoder improved the performance of the model under the 2-stage pre-training strategy.

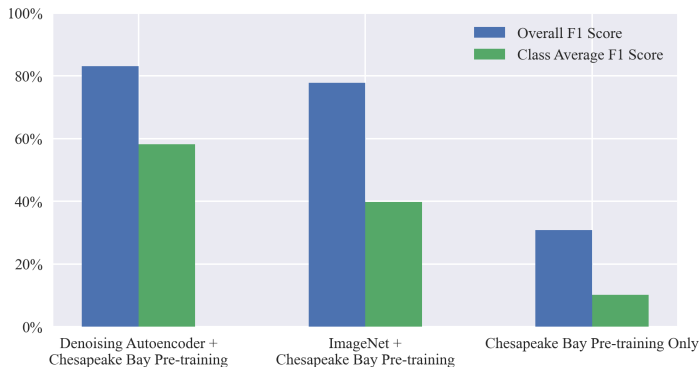


Figure 12: Performance of models on the Mississippi land cover dataset.

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Pre-training Strategies

An additional fully supervised pre-training stage substantially improves the performance of the model.

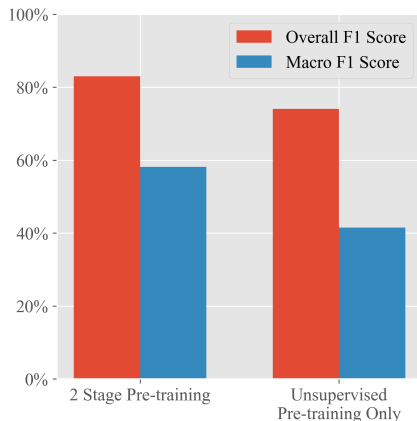


Figure 13: Performance of the best model with and without additional supervised pre-training.

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Figure 14: Confusion matrix of the final model on the Mississippi land cover dataset.

Mississippi Hi-Res Land Cover



Figure 15: 2023 Mississippi Land Cover Map.

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- Large scale pre-training of a deep model using an unsupervised regimen can enable the training of an over-parameterized deep model on a small amount of labeled data.

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- Large scale pre-training of a deep model using an unsupervised regimen can enable the training of an over-parameterized deep model on a small amount of labeled data.
- Further, additional pre-training using a modest amount of labeled data can boost the performance of the model even further.

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- Large scale pre-training of a deep model using an unsupervised regimen can enable the training of an over-parameterized deep model on a small amount of labeled data.
- Further, additional pre-training using a modest amount of labeled data can boost the performance of the model even further.
- Still, classifying more visually intricate land cover classes remains a challenge, even with a large model.

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- More data — clear need to improve the accuracy of cultivated crops and barren land classes.

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- Post-processing using OBIA-inspired techniques to refine classification map edges.

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- More data — clear need to improve the accuracy of cultivated crops and barren land classes.
- Post-processing using OBIA-inspired techniques to refine classification map edges.
- Applying model to super-resolved satellite imagery.

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